

Supplemental Online Content

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eMethods

eTable 1. Distribution of ALSFRS-R by visit and treatment group

eTable 2. Primary Efficacy Analysis of Function and Mortality with Bayesian Shared Parameter Model

eTable 3A. Results for secondary and exploratory outcomes using repeated-measures analysis using shared placebo participants (Full Analysis Set)

eTable 3B. Results for secondary and exploratory outcomes using repeated-measures analysis using only Regimen C (CNM-Au8) placebo participants (Efficacy Regimen Only dataset)

eTable 4. Subgroup Forest Plot for ALSFRS-R

eTable 5. Treatment-Emergent Adverse Events (TEAE) with $\geq 5\%$ incidence in either group

eFigure. Serum Neurofilament Sensitivity Analysis

This supplemental material has been provided by the authors to give readers additional information about their work.

eMethods

Trial Design and Oversight

The HEALEY ALS Platform Trial is a novel, efficient, perpetual design, platform trial designed to test multiple investigational products against placebo in parallel. In this platform trial, each new investigational product is tested as a regimen specific appendix, or “Regimen” to the master protocol. CNM-Au8 was evaluated in Regimen C (RGC) of the HEALEY ALS Platform Trial, a randomized, double-blind, placebo-controlled trial enrolled and followed participants at 54 Northeast ALS Consortium (NEALS) centers in the United States between July 2020 and March 2022. The trial was conducted in accordance with Good Clinical Practice guidelines of the International Conference on Harmonization and ethical principles of the Declaration of Helsinki. Protocol approval was provided for all trial sites by a central Institutional Review Board, the Massachusetts General Brigham Human Research Committee. All participants provided written informed consent prior to screening.

The trial was designed by and conducted through the NEALS network, a global collaborative trial network (www.neals.org), in collaboration with Clene Nanomedicine, Inc. (Clene). Participant-level data were obtained at each trial site and stored in the electronic data capture system for the trial at Massachusetts General Hospital. An independent Data and Safety Monitoring Board reviewed unblinded safety data at pre-defined intervals throughout the trial. Statistical analyses were performed by biostatisticians at Berry

Consultants, Massachusetts General Hospital, and Pentara Corporation. Clene provided the active drug and placebo and was involved in the trial design, data analysis, and manuscript review.

Trial Participants

Those who were eligible for the platform trial were adults with a diagnosis of clinically possible, probable, lab-supported probable, or definite ALS defined by revised El Escorial criteria, disease duration ≤ 36 months, a vital capacity $\geq 50\%$ predicted for age, height, and sex, the ability to swallow pills and liquids, and either no use of riluzole and/or edaravone or stable dosing of riluzole and/or edaravone for more than 30 days or one cycle, respectively. Those eligible for the CNM-Au8 regimen could not be allergic to gold. Full inclusion and exclusion criteria can be viewed in the trial protocol available as a Supplement.

Randomization techniques

Both the randomization to regimen, which was not blinded, and the randomization to active drug or placebo within a regimen, which was blinded to both participant and investigator, were done using a pre-determined randomization schedule. In order to ensure balance in regimens and treatment groups by current medication use, both randomization schedules were stratified by baseline use of riluzole and edaravone (4 strata total).

Random assignment to a regimen was done. Random assignment to a regimen within a stratum uses a block urn design (Zhao and Wang; Block urn design - a new randomization algorithm for sequential trials with two or more treatments and balanced or

unbalanced allocation. Contemp Clin Trials. 2011 Nov;32(6):953-61) with one instance of each regimen in a given block and modified to remove from consideration any regimens to which the participant cannot be assigned (termed "unavailable regimens").

Within each strata a blocking schema was used to create the randomization schedule. Blocks were of size 8 (3 for 30mg, 3 for 60mg, 2 for Placebo). Randomization schedules were created using SAS software and validated to ensure no discernable patterns were seen before implementation.

Trial Interventions and Procedures

Eligible participants were randomized 3:3:2 ratio to receive CNM-Au8 60mg daily, CNM-Au8 30mg daily, or matching placebo, with stratification for edaravone and/or riluzole use. Trial duration was 24 weeks. Active drug and placebo were provided as liquid in individual containers. Participants were instructed to take two 60mL bottles of study drug or placebo daily in the morning at least 30 minutes prior to food intake. For the 60mg dose, each bottle contained 30mg of CNM-Au8. For the 30mg dose, each bottle contained 15mg CNM-Au8. The primary analysis compared those receiving active CNM-Au8, at either the 60mg or 30mg daily dosage, to placebo. Clinic visits were conducted at baseline, Weeks 4, 8, 16 and 24, and telephone visits at Weeks 2, 12, and 20, with a final telephone follow-up at week 28. Participants who completed the randomized, double-blind trial were eligible for enrollment in an open-label extension (OLE) evaluating the long-term effects of CNM-Au8 (NCT04414345).

Outcomes

The primary endpoint was change in disease severity from baseline through 24 weeks as measured by the ALS Functional Rating Scale revised (ALSFRS-R) total score and survival and analyzed using a Bayesian shared-parameter model (see Primary Analysis Methods below). Survival and PAV status were recorded during planned 24-week double-blind follow-up and censored at the last date known alive if prior to the end of the week 24 visit window.

The three secondary clinical efficacy outcomes (in hierarchical order) included 1) the Combined Assessment of Function and Survival (CAFS) model, a joint-rank test combining ALSFRS-R decline and survival free of permanent assisted ventilation (PAV), defined as more than 22 hours daily for more than 7 days; 2) rate of decline in percent predicted of normal (PPN) slow vital capacity (SVC) and; 3) survival that is free of PAV.

Safety was assessed via documentation of treatment-emergent adverse events (TEAEs) at each trial visit. Symptoms of ALS progression, including those consistent with disease progression, were recorded as TEAEs. Any worsening of a disease progression measure that was being recorded and analyzed separately (i.e., ALSFRS-R or SVC) was not recorded as a TEAE. Safety was additionally assessed based on laboratory assays, vital signs, electrocardiograms, and the Columbia Suicide Severity Rating Scale (C-SSRS).

Pre-specified exploratory measures included: 4 biofluid biomarkers (serum and CSF neurofilament light (NfL), serum creatinine,

urinary p75 extracellular domain (p75ecd), ; 6 clinical outcome measures (HHD), quantitative motor speech (9 features), home vital capacity, ALSFRS-R domain scores, the ALS Assessment Questionnaire (ALSAQ-40) and its domain scores, and the Center for Neurologic Study – Bulbar Function Scale (CNS-BFS); and 11 time-to-event analyses (3 events related to cutoffs in trial endpoints, such as time to decline of at least 6 points on the ALSFRS-R, and 8 exploring clinical events such as time to first serious adverse event). Plasma NfL was added later as a post-hoc analysis. Only relevant exploratory analyses are presented in this manuscript. Details about all exploratory measures and their analyses can be found in the statistical analysis plan (available as a Supplement).

Analysis of Dosages

The main analysis for primary, secondary, safety and exploratory measures was on the combined 30mg and 60mg daily dosages compared to placebo. Prespecified exploratory analyses compared each dose to placebo directly for all primary, secondary, and exploratory endpoints.

Analysis Populations

The primary analysis population for both the primary and secondary endpoints, referred to as the Full Analysis Set (FAS), included all participants randomized to the Regimen C active treatment arm, all participants randomized to the placebo arm within Regimen C, and all placebo participants from Regimens A, B, and D. Participants were analyzed according to the treatment group to which they were randomized within their respective regimen to estimate a treatment policy estimand. Additional prespecified efficacy analyses were

performed using only Regimen C participants (CNM-Au8 and placebo), called the regimen-only analyses. Safety analyses were performed in an equivalent full analysis set for safety that included all Regimen C participants who initiated treatment and all placebo participants from Regimens A, B, and D except those who were known to be ineligible for the CNM-Au8 regimen (i.e., those with gold allergy).

The primary analysis on the FAS used longitudinal ALSFRS-R scores from all scheduled visits (Weeks 0, 4, 8, 12, 16, 20 and 24) or from any early termination or ad hoc visits substituting for a planned visit.

Primary analysis methods

The primary analysis is a Bayesian shared parameter model of function and survival that provides an integrated estimate of the increase or decrease in the rate of disease progression on treatment relative to control. The analysis model has components for each endpoint that are linked through an integrated estimate of disease slowing in treatment relative to control across the two endpoints (denoted as the disease rate ratio, or DRR). This composite variable strategy fully incorporates effects of treatment on the intercurrent event of death. The DRR measures the ratio of ALSFRS-R slopes within the functional component or the hazard ratio within the survival component with values less than 1 indicating slowing in disease progression. This parameter can be interpreted as the slowing in the rate of disease progression measured across both endpoints. A DRR of 0.50 corresponds to a 50% slowing in the rate of disease progression. If a participant in the control group would be expected to decrease 6 points over 6 months, then under a DRR

of 0.50, they would be expected to reach the same milestone (ALSFRS-R decrease of 6 points) over 12 months on treatment.

Similarly for survival, if they were expected to survive for a median time of 9 months on control, under the treatment they would be expected to survive for 18 months. In the functional component, ALSFRS-R is measured for those who have survived using a linear repeated measure model that adjusts for baseline covariates and accommodates potential differences in the shared control across regimens. The survival component is measured through an exponential proportional hazards model. The shared treatment effect between the ALSFRS-R and mortality components allows the participants who were lost to follow-up due to mortality to inform treatment effect estimates beyond their censored ALSFRS-R longitudinal data. The degree to which treatment effects on mortality inform the shared-treatment effect parameter depends on the mortality rate within the study.

The primary analysis includes concurrent shared controls across regimens. It is expected that there will not be systematic differences in rates of progression in controls across regimens. The platform enrolls participants under the same master protocol, across the same sites, during the same time period, and with only very minor differences in regimen-specific inclusion/exclusion. However, minor differences across the shared controls may be present due to these minor differences in inclusion/exclusion and due to differences in the modes of administration. To account for these potential differences, the primary analysis is conducted within a Bayesian hierarchical meta-analytic framework. As such, the regimen-specific random effects on the rate of progression for controls are assumed to come from an underlying hierarchical distribution with unknown population mean and variability. The population mean and variability (hyper-parameters) within the hierarchical distribution are given a prior distribution, and posterior estimates are obtained from the data. If the variability in the rate of progression is estimated to be small, the model will borrow more information

across regimens and the analysis will be similar to a pooled analysis across regimens. If the variability in the rate of progression is estimated to be large across regimens, the model will borrow less across regimens and the analysis will be similar to an independent analysis across regimens.

The Bayesian posterior distribution of all model parameters is calculated using Markov chain Monte Carlo (MCMC). We drew at least 500,000 samples from the posterior distribution with a thinning of 10. The exact number of samples from the MCMC was determined based on convergence diagnostics to ensure an adequate effective sample size of the posterior distribution.

Prespecified sensitivity analyses, including CAFS, which was employed as a sensitivity analysis, in addition to a secondary endpoint, were performed to assess the sensitivity of the results to the primary analysis population, modeling assumptions, and missing data assumptions as described in Statistical Analysis Plan (available as a Supplement).

Interim Analyses

A regimen could be stopped early should prespecified interim analyses indicate futility. Interim analyses occurred simultaneously for all actively enrolling regimens and included any regimen within the master protocol that had 40 active randomized participants (30 study drug and 10 placebo). Subsequent interim analyses occurred every 12 weeks. As described in the master protocol, early futility would be declared if the regimen had a posterior probability of <5% that CNM-Au8 would slow ALSFRS-R decline by at least 10%. There was no stopping rule for early success.

Secondary efficacy analysis methods

SVC was assessed using both a repeated measures linear mixed model and random slopes model with the following covariates, centered to the analysis population: months since symptom onset, pre-baseline change in ALSFRS-R slope (delta-FRS), baseline riluzole and edaravone use, and their interaction with time. Models run on the full analysis set, which includes shared placebos from other regimens, random regimen-specific intercepts are included.

Each participant's vital status was verified at the time of the last patient last visit in the RCT. Rates of death, death-equivalent events (permanent assisted ventilation (PAV) defined as use of ventilation for >22 hours per day for more than 7 days in a row), and hospitalization were analyzed using a Cox proportional hazards model, with covariates of pre-baseline ALSFRS-R slope, months since symptom onset, riluzole and edaravone use at baseline, and age at baseline. A separate supportive analysis was run which included centered baseline serum NfL level and the interaction of centered baseline serum NfL level and visit as additional covariates.

Exploratory Analyses

As a part of the Healey Platform Trial Master Protocol, serum was analyzed using single molecule array (Simoa) testing, a digital immunoassay (Quanterix) as a single-plex assay. Two replicates from each aliquot were performed. If the sample NfL was below the

limit of detection or only one replicate yielded an NfL concentration, the replicates were repeated. If the coefficient of variation (CV) for these two replicates was >25%, the replicates were repeated; the repeat replicates were used unless their CV was higher than the original. If a serum sample value was <LOQ, the value was imputed by using the assay LOD. For the primary analysis, no outliers were omitted.

Serum NfL in the Full Analysis set was the prespecified exploratory endpoint for NfL. However, for this regimen, serum samples were tested with all longitudinal samples from a single participant tested on the same plate (different than other regimens). Only regimen samples were tested in this way, thus a post-hoc exploratory analysis of only regimen samples is presented. In addition, plasma samples were tested using a multiplex assay (Neurology 4 Plex-A, Quanterix) that included NfL, glial fibrillary acidic protein (GFAP), tau, and ubiquitin carboxy-terminal hydrolase L1 (UCH-L1) with two replicates from each aliquot performed. The plasma analysis was pre-specified in Clene Nanomedicine's Pharmacodynamic SAP originally as UCH-L1 alone, but due to non-availability of the UCH-L1 assay the Neurology 4 Plex-A (including NfL) was analyzed, thus it is considered post-hoc for the purposes of this manuscript.

Sites were instructed to process serum and plasma samples within 30 minutes of collection and stored at -80C for batch shipment to the lab using standard operating procedures as described in prior studies. Samples were tested on a 96-well plate with high and low-level quality control standards to allow for correction of plate-to-plate variability. NfL values were log-transformed prior to analysis.

Data was analyzed by mixed model repeat measures (MMRM) with prespecified covariates including pre-treatment ALSFRS-R slope (delta-FRS), months from symptom onset, use of riluzole, use of edaravone; covariate by visit interaction, and treatment by visit interaction. NfL was analyzed in the regimen only dataset. For the primary analyses of neurofilament, all values obtained were included in the analysis (i.e., no outliers were excluded). To assess whether the serum NfL result was sensitive to outliers, a post hoc analysis was performed that excluded outliers (observations with very low NfL values, namely those <4 pg/mL or <2% of the maximum for a given participant. This excluded one baseline NfL for one participant in the placebo group.

Urinary p75 extracellular domain (p75ecd) was analyzed at Flinders University in the Rogers Lab. Urinary p75 extracellular domain was quantified using a sandwich enzyme-linked immunosorbent assay (ELISA) in 96-well plates, as previously described. Only samples from the CNM-Au8 regimen were analyzed for urinary p75ecd.

Prespecified exploratory outcomes for time to event analyses included time to hospitalization, placement of a feeding tube, use of assisted ventilation, King's Stage 4, death alone, and death or PAV. Analysis used an adjusted Cox regression using the same covariates described for the primary endpoint as well as participant age. King's stage analysis was interval censored; other variables were right censored. Death or PAV was censored at the time the participant was last known to be alive.

Subgroup Analyses

Differences in treatment efficacy were explored in several prespecified subgroups: baseline riluzole use (yes vs. no), and baseline edaravone use (yes vs. no), age (<65 years vs 65 or older), sex, race, ethnicity, weight categories, BMI categories, chronic kidney disease stage, time since onset of weakness (<18 months vs 18 months or more), El Escorial criteria with time since onset of weakness, baseline symptom severity (all ALSFRS-R questions scored 2 or greater vs any scored 0 or 1), early disease state, baseline serum urate concentration (<5.5mg/dL vs 5.5mg/dL or greater), baseline serum NfL concentration (median split), site of onset (any bulbar vs none), pre-baseline rate of progression by delta-FRS (fast vs. slow with median split), and clinic site (sites with <5 participants were pooled). Subgroup specific estimates will be obtained by including subgroup, subgroup $\times$ time, and subgroup $\times$ treatment $\times$ time terms to a random-slope model for primary and secondary efficacy endpoints using the FAS population.

eTable 1. Distribution of ALSFRS-R by visit and treatment group

Visit	CNM-Au8 pooled		Shared Placebo		Regimen Placebo	
	n	mean (SD)	n	mean (SD)	n	mean (SD)
Baseline	120	34.3 (6.6)	164	35.1 (6.7)	41	36.1 (5.9)
Week 4	120	33.0 (7.0)	160	34.5 (7.0)	40	35.4 (6.7)
Week 8	115	32.3 (7.3)	155	33.4 (7.4)	38	34.3 (7.1)
Week 12	115	31.4 (7.7)	152	32.7 (7.9)	39	33.7 (7.9)
Week 16	110	30.5 (7.7)	148	31.8 (8.3)	36	32.5 (7.5)
Week 20	111	29.7 (7.8)	143	31.2 (8.6)	36	32.6 (7.9)
Week 24	111	29.1 (7.9)	143	30.2 (8.7)	35	31.4 (8.2)

eTable 2. Primary Efficacy Analysis of Function and Mortality with Bayesian Shared Parameter Model

Parameter	Median	Mean (SD)	95% Credible Interval	Pr(DRR < 1.0)	Pr(DRR < 0.9)
DRR					
Function and Mortality	0.96	0.97 (0.099)	(0.783, 1.175)	0.6450	0.2505
ALSFRS-R Slopes (points per month)					
Regimen C placebo w/ sharing	−1.03	−1.03 (0.073)	(−1.181, −0.894)		
Pooled CNM-Au8	−1.00	−1.00 (0.075)	(−1.143, −0.847)		
Mortality Event Rate (events per month)					
Shared placebo	0.010	0.010 (0.0026)	(0.0054, 0.0154)		
Pooled CNM-Au8	0.009	0.009 (0.0025)	(0.0052, 0.0150)		
Covariates					
Months since onset	1.03	1.03 (0.046)	(0.938, 1.120)		
Pre-baseline Slope	1.39	1.40 (0.079)	(1.255, 1.565)		
Edaravone Use	1.13	1.13 (0.101)	(0.949, 1.344)		
Riluzole Use	1.03	1.03 (0.093)	(0.854, 1.220)		

Abbreviations: DRR = Disease Rate Ratio

Note: The Bayesian shared parameter model includes covariates for baseline use of Edaravone, baseline use of Riluzole, months since onset of symptoms, and pre-baseline slope of ALSFRS-R.

eTable 3A. Results for secondary and exploratory outcomes using repeated-measures analysis using shared placebo participants (Full Analysis Set)

FAS Population		24-week Change Estimate		Difference from Shared Placebo	
Endpoint	CNM-Au8 Pooled (n=120)	Shared Placebo (n=164)	Difference (SE)	95% CI	Pvalue
ALSFRS-R total Score	-5.47 (0.43)	-5.51 (0.36)	0.0 (0.55)	-1.03, 1.11	0.94
SVC (% predicted)	-9.32 (1.36)	-8.53 (1.15)	-0.78 (1.77)	-4.25, 2.68	0.66
Dose Specific	CNM-Au8 30 mg (n=59)	Shared Placebo (n=164)	Difference (SE)	95% CI	P value
ALSFRS-R total Score	-5.70 (0.59)	-5.51 (0.36)	-0.19 (0.69)	-1.53, 1.16	0.79
SVC (% predicted)	-7.84 (1.84)	-8.53 (1.15)	0.69 (2.19)	-3.55, 4.92	0.75
	CNM-Au8 60 mg (n=61)	Shared Placebo (n=164)	Difference (SE)	95% CI	P value
ALSFRS-R total Score	-5.24 (0.59)	-5.51 (0.36)	0.27 (0.68)	-1.07, 1.60	0.69
SVC (% predicted)	-10.79 (1.99)	-8.53 (1.15)	-2.26 (2.29)	-6.75, 2.24	0.33

eTable 3B. Results for secondary and exploratory outcomes using repeated-measures analysis using only Regimen C (CNM-Au8) placebo participants (Efficacy Regimen Only dataset)

ERO Population		24-week Change Estimate		Difference from Regimen Placebo	
Endpoint	CNM-Au8 Pooled (n=120)	Regimen Placebo (n=41)	Difference (SE)	95% CI	P value
ALSFRS-R total Score	-5.45 (0.44)	-5.39 (0.71)	-0.07 (0.83)	-1.71, 1.57	0.94
SVC (% predicted)	-9.57 (1.38)	-6.74 (2.29)	-2.83 (2.67)	-8.10, 2.44	0.29
Plasma NfL (% change)	-2.3	7.9	-9.5	-18.0, 0	0.04
Serum NfL (% change)	0.4	26.8	-26.4	-50.3, -2.6	0.03
Dose Specific					
	CNM-Au8 30 mg (n=59)	Regimen Placebo (n=41)	Difference (SE)	95% CI	P value
ALSFRS-R total Score	-5.56 (0.60)	-5.39 (0.71)	-0.17 (0.93)	-2.00, 1.66	0.85
SVC (% predicted)	-7.61 (1.86)	-6.74 (2.29)	-0.87 (2.94)	-6.69, 4.95	0.77
Plasma NfL (% change)	-1.3	7.9	-8.5	-18, 2	0.10
Serum NfL (% change)	1.4	26.8	-25.5	-52.2, 1.3	0.06
	CNM-Au8 60 mg (n=61)	Regimen Placebo (n=41)	Difference (SE)	95% CI	P value
ALSFRS-R total Score	-5.35 (0.61)	-5.39 (0.71)	0.04 (0.93)	-1.79, 1.87	0.97
SVC (% predicted)	-11.53 (1.98)	-6.74 (2.29)	-4.79 (3.03)	-10.76, 1.19	0.12
Plasma NfL (% change)	-3.4	7.9	-10.5	-20, 0	0.04
Serum NfL (% change)	-0.5	26.8	-27.4	-45.1, -0.7	0.05

FAS = Full Analysis Set (includes shared placebos), ERO = Efficacy Regimen Only (includes only regimen-specific placebos),

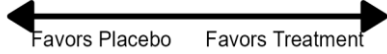
ALSFRS-R = Amyotrophic Lateral Sclerosis Functional Rating Scale – Revised; SVC = Slow Vital Capacity; CI = confidence interval; SE = standard error; NfL = Neurofilament Light; GMR – geometric mean ratio

*Serum and plasma NfL was modeled as log-transformed and exponentiated back to the original scale to present results.

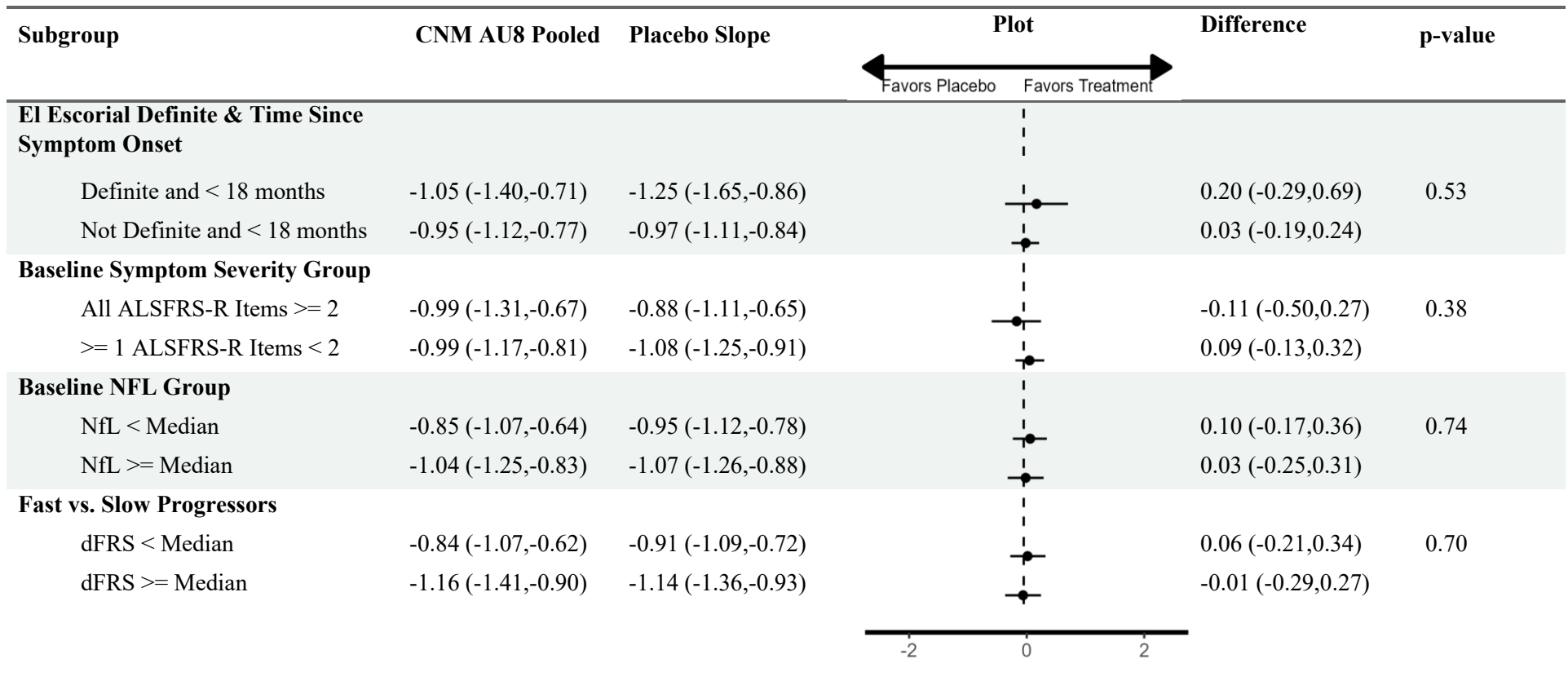
Models adjusted for time since symptom onset, pre-baseline ALSFRS-R slope, edaravone use at baseline, riluzole use at baseline and their interactions with visit.

eTable 4. Subgroup Forest Plot for ALSFRS-R

This forest plot shows the behavior of the Amyotrophic Lateral Sclerosis Functional Rating Scale – Revised (ALSFRS-R) in multiple subgroups, comparing the combined (pooled) CNM-Au8 arm to the shared placebo arm using a linear mixed model approach as described in the methods section. For each subgroup, the difference between the estimated slopes for the CNM-Au8 arm and the placebo arm is shown as a point estimate on the plot with 95% confidence intervals (95%CI) represented as the horizontal solid line. The vertical dashed line represents “no difference.” In every subgroup analyzed, the 95% CI crosses the “no difference” line, indicating that no subgroups showed a statistically significant benefit from CNM-Au8 on the slope of ALSFRS-R decline.

Subgroup	CNM AU8 Pooled	Placebo Slope	Plot	Difference	p-value
					
All Participants	-0.98 (-1.13,-0.82)	-1.01 (-1.13,-0.88)		0.03 (-0.16,0.23)	
Riluzole Use					
N	-1.07 (-1.39,-0.75)	-0.88 (-1.15,-0.61)		-0.19 (-0.60,0.22)	0.22
Y	-0.95 (-1.12,-0.78)	-1.05 (-1.19,-0.90)		0.10 (-0.12,0.32)	
Edaravone Use					
N	-1.00 (-1.17,-0.83)	-0.96 (-1.10,-0.81)		-0.04 (-0.27,0.18)	0.19
Y	-0.91 (-1.22,-0.60)	-1.17 (-1.43,-0.90)		0.26 (-0.14,0.66)	
Riluzole & Edaravone Use					
Neither	-1.15 (-1.49,-0.81)	-0.82 (-1.10,-0.54)		-0.33 (-0.77,0.10)	
Riluzole Only	-0.94 (-1.14,-0.75)	-1.01 (-1.18,-0.83)		0.06 (-0.20,0.32)	
Edaravone Only	-0.49 (-1.45,0.47)	-1.38 (-2.27,-0.49)		0.89 (-0.42,2.20)	
Both	-0.96 (-1.28,-0.64)	-1.15 (-1.43,-0.88)		0.19 (-0.23,0.61)	
Age Group					
< 65	-0.97 (-1.15,-0.80)	-1.03 (-1.18,-0.88)		0.05 (-0.18,0.28)	0.81
>= 65	-0.95 (-1.23,-0.68)	-0.95 (-1.19,-0.71)		0.00 (-0.36,0.36)	

Subgroup	CNM AU8 Pooled	Placebo Slope	Plot	Difference	p-value
			Favors Placebo Favors Treatment		
Sex					
F	-0.98 (-1.22,-0.74)	-1.10 (-1.33,-0.87)			



eTable 5. Treatment-Emergent Adverse Events (TEAE) with $\geq 5\%$ incidence in either group

Treatment Emergent Adverse Events (TEAE)	Pooled CNM-Au8	CNM-Au8 30mg	CNM-Au8 60mg	Shared Placebo	Regimen-Specific Placebo
	(N=120)	(N=59)	(N=61)	(N=163)	(N=41)
MedDRA SOC Preferred Term	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events
Any TEAE	111 (92.5%), 723	54 (91.5%), 293	57 (93.4%), 430	146 (89.6%), 889	38 (92.7%), 228
Mild	41 (34.2%), 500	21 (35.6%), 205	20 (32.8%), 295	68 (41.7%), 673	13 (31.7%), 151
Moderate	44 (36.7%), 165	22 (37.3%), 66	24 (39.3%), 99	53 (32.5%), 164	15 (36.6%), 61
Severe	24 (20%), 58	11 (18.6%), 22	13 (21.3%), 36	25 (15.3%), 47	10 (24.4%), 16
Serious TEAEs	16 (13.3%), 20	6 (10.2%), 9	10 (16.4%), 11	15 (9.2%), 21	7 (17.1%), 12
Treatment-Related Serious TEAE	0 (0.0%), 0	0 (0.0%), 0	0 (0.0%), 0	2 (1.2%), 3	1 (2.4%), 2
Deaths	4 (3.3%), 4	1 (1.7%), 1	3 (4.9%), 3	4 (2.5%), 4	2 (4.9%), 2
TEAEs Leading to Study Drug Withdrawal	8 (6.7%), 8	4 (6.8%), 4	4 (6.6%), 4	11 (6.7%), 17	3 (7.3%), 4
TEAEs Leading to Study Drug Interruption	9 (7.5%), 15	6 (10.2%), 11	3 (4.9%), 4	12 (7.4%), 24	4 (9.8%), 6

Treatment Emergent Adverse Events (TEAE)	Pooled CNM-Au8	CNM-Au8 30mg	CNM-Au8 60mg	Shared Placebo	Regimen-Specific Placebo
	(N=120)	(N=59)	(N=61)	(N=163)	(N=41)
MedDRA SOC Preferred Term	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events
TEAEs Leading to Study Drug Reduction	3 (2.5%), 5	0 (0.0%), 0	3 (4.9%), 5	5 (3.1%), 12	1 (2.4%), 1
Fall	26 (21.7%), 44	13 (22.0%), 22	13 (21.3%), 22	43 (26.4%), 79	13 (31.7%), 17
Muscular weakness	24 (20.0%), 39	10 (16.9%), 18	14 (23.0%), 21	45 (27.6%), 67	12 (29.3%), 17
Diarrhoea	23 (19.2%), 33	9 (15.3%), 12	14 (23.0%), 21	12 (7.4%), 14	4 (9.8%), 5
Nausea	17 (14.2%), 21	9 (15.3%), 10	8 (13.1%), 11	14 (8.6%), 16	6 (14.6%), 8
Fatigue	13 (10.8%), 17	6 (10.2%), 8	7 (11.5%), 9	30 (18.4%), 33	9 (22.0%), 9
Headache	17 (14.2%), 26	9 (15.3%), 17	8 (13.1%), 9	19 (11.7%), 23	4 (9.8%), 6
Neuromyopathy	14 (11.7%), 21	6 (10.2%), 8	8 (13.1%), 13	22 (13.5%), 33	6 (14.6%), 7
Constipation	17 (14.2%), 19	4 (6.8%), 4	13 (21.3%), 15	20 (12.3%), 24	3 (7.3%), 4
Dyspnoea	13 (10.8%), 14	7 (11.9%), 8	6 (9.8%), 6	10 (6.1%), 11	4 (9.8%), 5
Dysarthria	15 (12.5%), 16	9 (15.3%), 10	6 (9.8%), 6	11 (6.7%), 13	2 (4.9%), 2
Dysphagia	10 (8.3%), 10	6 (10.2%), 6	4 (6.6%), 4	18 (11.0%), 22	5 (12.2%), 6
Oedema peripheral	7 (5.8%), 13	2 (3.4%), 3	5 (8.2%), 10	12 (7.4%), 15	4 (9.8%), 4
Tension headache	7 (5.8%), 10	1 (1.7%), 1	6 (9.8%), 9	10 (6.1%), 15	4 (9.8%), 4

Treatment Emergent Adverse Events (TEAE)	Pooled CNM-Au8	CNM-Au8 30mg	CNM-Au8 60mg	Shared Placebo	Regimen-Specific Placebo
	(N=120)	(N=59)	(N=61)	(N=163)	(N=41)
MedDRA SOC Preferred Term	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events
Urinary tract infection	10 (8.3%), 13	4 (6.8%), 6	6 (9.8%), 7	10 (6.1%), 10	1 (2.4%), 1
Arthralgia	6 (5.0%), 12	3 (5.1%), 8	3 (4.9%), 4	6 (3.7%), 6	3 (7.3%), 3
Vomiting	6 (5.0%), 9	4 (6.8%), 5	2 (3.3%), 4	6 (3.7%), 6	2 (4.9%), 2
Amyotrophic lateral sclerosis	8 (6.7%), 10	2 (3.4%), 2	6 (9.8%), 8	1 (0.6%), 1	0 (0.0%), 0
Skin laceration	6 (5.0%), 7	2 (3.4%), 2	4 (6.6%), 5	4 (2.5%), 8	2 (4.9%), 2
Cough	7 (5.8%), 7	1 (1.7%), 1	6 (9.8%), 6	4 (2.5%), 4	1 (2.4%), 1
Decreased appetite	5 (4.2%), 5	2 (3.4%), 2	3 (4.9%), 3	12 (7.4%), 13	3 (7.3%), 3
Insomnia	5 (4.2%), 6	0 (0.0%), 0	5 (8.2%), 6	2 (1.2%), 2	2 (4.9%), 2
Salivary hypersecretion	5 (4.2%), 6	1 (1.7%), 1	4 (6.6%), 5	9 (5.5%), 9	2 (4.9%), 2
Depression	3 (2.5%), 3	1 (1.7%), 1	2 (3.3%), 2	8 (4.9%), 8	4 (9.8%), 4
Faeces discoloured	7 (5.8%), 7	1 (1.7%), 1	6 (9.8%), 6	0 (0.0%), 0	0 (0.0%), 0
Post-traumatic pain	4 (3.3%), 4	2 (3.4%), 2	2 (3.3%), 2	6 (3.7%), 6	3 (7.3%), 3
Weight decreased	2 (1.7%), 2	0 (0.0%), 0	2 (3.3%), 2	5 (3.1%), 5	5 (12.2%), 5
Pain in extremity	5 (4.2%), 7	1 (1.7%), 1	4 (6.6%), 6	9 (5.5%), 10	1 (2.4%), 1

Treatment Emergent Adverse Events (TEAE)	Pooled CNM-Au8	CNM-Au8 30mg	CNM-Au8 60mg	Shared Placebo	Regimen-Specific Placebo
	(N=120)	(N=59)	(N=61)	(N=163)	(N=41)
MedDRA SOC Preferred Term	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events	Number of Participants (%), Number of events
Pruritus	6 (5.0%), 8	0 (0.0%), 0	6 (9.8%), 8	4 (2.5%), 6	0 (0.0%), 0
Dizziness	4 (3.3%), 4	3 (5.1%), 3	1 (1.6%), 1	10 (6.1%), 13	2 (4.9%), 3
Increased upper airway secretion	5 (4.2%), 6	3 (5.1%), 4	2 (3.3%), 2	4 (2.5%), 4	1 (2.4%), 1
Anxiety	4 (3.3%), 4	0 (0.0%), 0	4 (6.6%), 4	8 (4.9%), 8	2 (4.9%), 2
COVID-19	3 (2.5%), 3	2 (3.4%), 2	1 (1.6%), 1	11 (6.7%), 11	3 (7.3%), 3
Pulmonary embolism	5 (4.2%), 5	1 (1.7%), 1	4 (6.6%), 4	2 (1.2%), 2	1 (2.4%), 1
Sinusitis	6 (5.0%), 6	4 (6.8%), 4	2 (3.3%), 2	0 (0.0%), 0	0 (0.0%), 0
Asthenia	4 (3.3%), 5	0 (0.0%), 0	4 (6.6%), 5	2 (1.2%), 2	1 (2.4%), 1
Back pain	5 (4.2%), 6	3 (5.1%), 4	2 (3.3%), 2	4 (2.5%), 10	0 (0.0%), 0
Flatulence	5 (4.2%), 5	0 (0.0%), 0	5 (8.2%), 5	0 (0.0%), 0	0 (0.0%), 0
Complication associated with device	3 (2.5%), 4	3 (5.1%), 4	0 (0.0%), 0	3 (1.8%), 3	0 (0.0%), 0
Alanine aminotransferase increased	3 (2.5%), 3	3 (5.1%), 3	0 (0.0%), 0	2 (1.2%), 2	0 (0.0%), 0
Aspartate aminotransferase increased	3 (2.5%), 3	3 (5.1%), 3	0 (0.0%), 0	1 (0.6%), 1	0 (0.0%), 0

eFigure. Serum Neurofilament Sensitivity Analysis

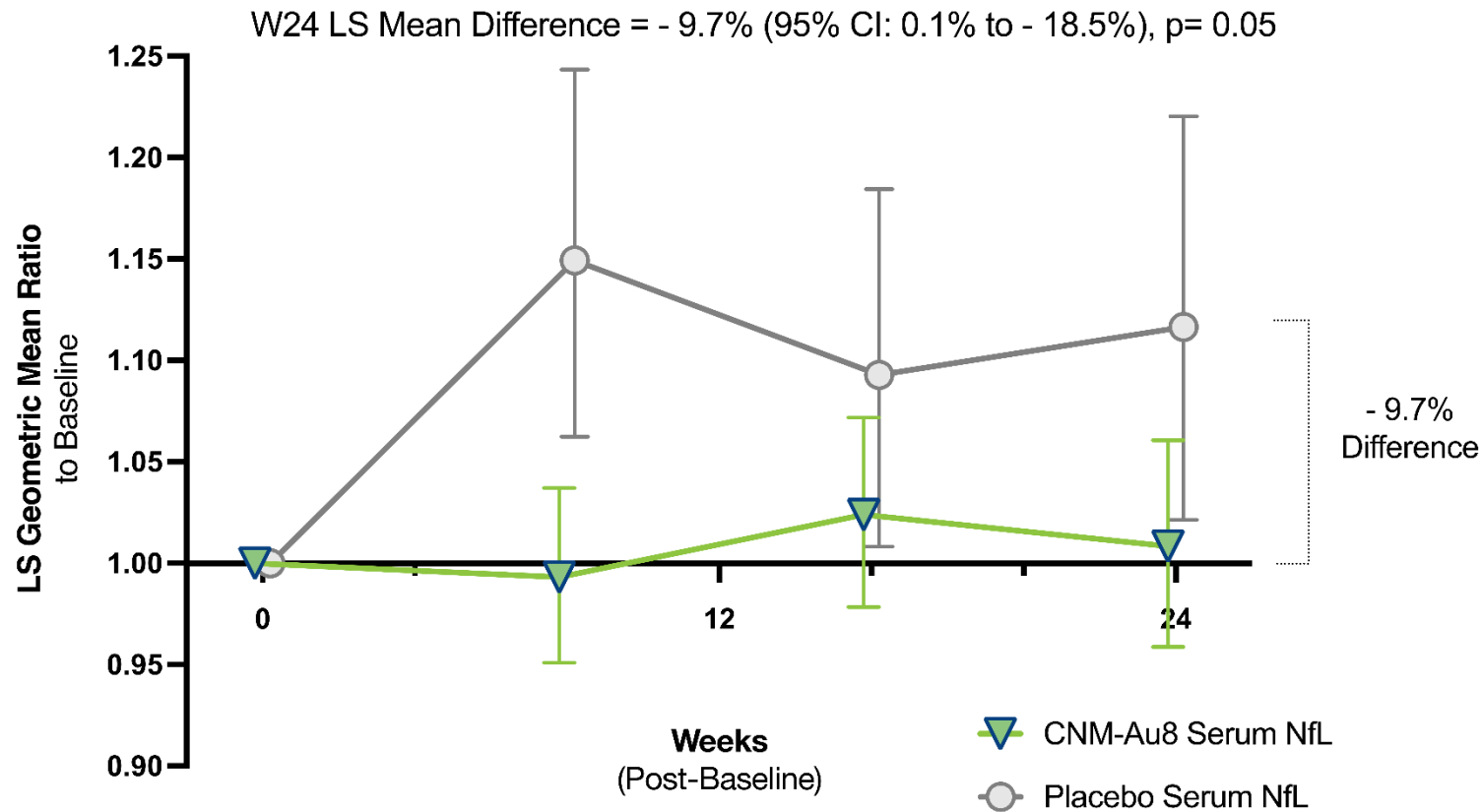
Sensitivity analysis of NfL tested in serum samples longitudinally at Baseline, Week 4 (plasma only), Week 8 (serum only), Week 16, and Week 24. NfL values <4pg/mL or <2% of the maximum for a given individual were excluded. This removed one baseline measure for one participant. Values were converted to natural log (Ln) scale prior to analysis and difference of least square means is shown at each timepoint. After back-transformation to compare values in the placebo and combined dosage CNM-Au8 treatment groups in this sensitivity analysis, the NfL levels were increased in the regimen placebo group (+11.6) and in the CNM-Au8 treatment group (+0.8%). The difference in change of the geometric mean comparing placebo to CNM-Au8 treatment groups was +9.7% (95% CI: -0.1% to +18.5%, p=0.051).

Serum Neurofilament Light Chain (NfL) [Sensitivity Analysis¹]

Regimen C Within Regimen Analysis | Quanterix 4NPA

All RCG Participants (n=161)

LS Means on Ln Scale $\pm$ 95% CI (Exponentiated)



¹ Excludes one placebo baseline value as a potential outlier